

Stellar PSFs Fake Hosts in Little Red Dots

Broad-band point-spread functions built from field stars systematically misassign light in high-redshift compact sources.

arXiv:2608.30459 — Source-conditioned polychromatic PSFs for compact high-redshift nuclei

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Background

Compact high-redshift sources are modelled by subtracting a point-spread function built from nearby field stars. This assumes the target's spectrum matches the stars used as reference.

No test existed for whether spectrally mismatched field stars bias broad-band host measurements of little red dots.

What this paper is selling

The authors derive the physical and mathematical origin of this colour-dependent PSF mismatch and demonstrate a correction. Applied to LRDs, it removes spurious extended flux and revises inferred sizes, luminosities and masses.

Key results to remember

- A pure point source yields wavelength-dependent residuals mistaken for a Sérsic host.
- Every tested LRD spectrum, stellar reference and detector position exceeds the published 11.8% F444W component.
- The bias inflates host flux, sizes and stellar masses for compact distant sources.

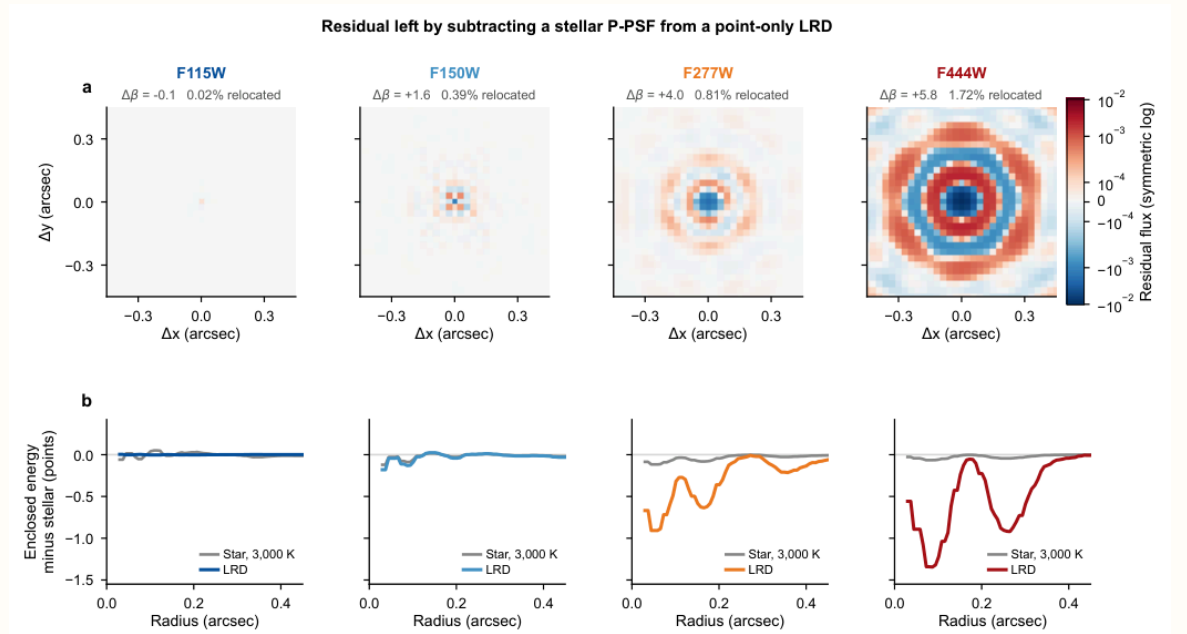


Figure 1 | Subtracting a stellar PSF from a point-only LRD leaves filter-dependent residuals that mimic a Sérsic host.

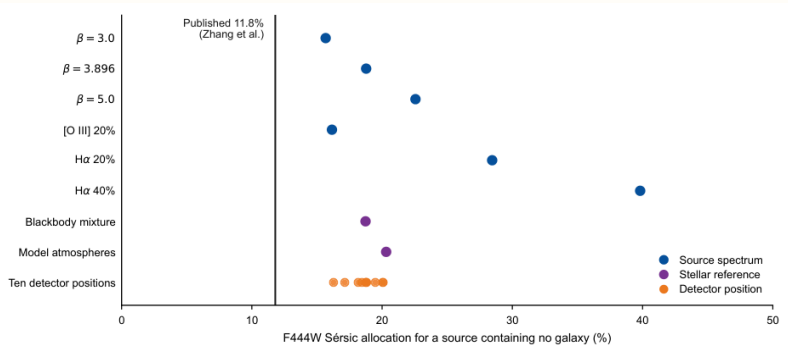


Figure 2 | Across all tested spectra, references and detector positions, the spurious F444W host allocation always exceeds the published 11.8%.